



**SYDNEY BOYS HIGH SCHOOL**  
MOORE PARK, SURRY HILLS

**2006**  
**TRIAL HIGHER SCHOOL**  
**CERTIFICATE**

# Mathematics      Extension 2

## *General Instructions*

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using black or blue pen.
- Board approved calculators may be used.
- All *necessary* working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Hand in your answer booklets in 3 sections.  
Section A (Questions 1 - 2),  
Section B (Questions 3 - 4)  
Section C (Questions 5 - 6)  
Section D (Questions 7 - 8).
- Start each **NEW** section in a separate answer booklet.

## **Total Marks - 120 Marks**

- Attempt Sections A - D
- All questions are of equal value.

Examiner:    *E. Choy*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

**Total marks – 120**  
**Attempt Questions 1 - 8**  
**All questions are of equal value**

Answer each section in a SEPARATE writing booklet. Extra writing booklets are available.

**SECTION A (Use a SEPARATE writing booklet)**

| Question 1 (15 marks)                                                  | Marks |
|------------------------------------------------------------------------|-------|
| (a) By first completing the square, evaluate the following integrals   |       |
| (i) $\int_{-1}^0 \frac{dx}{\sqrt{3-2x-x^2}}$                           | 2     |
| (ii) $\int_0^1 \sqrt{x(1-x)} dx$                                       | 2     |
| (b) Integrate the expressions below                                    |       |
| (i) $\int \frac{1}{x \ln x} dx$                                        | 1     |
| (ii) $\int x \ln x dx$                                                 | 2     |
| (iii) $\int \frac{x+1}{x^2+x+1} dx$                                    | 2     |
| (c) Use the technique of <i>integration by parts</i> to evaluate       | 2     |
| $\int_0^{\frac{1}{2}} \cos^{-1} x dx$                                  |       |
| (d) (i) Find real numbers $A$ , $B$ , and $C$ so that                  | 2     |
| $\frac{10}{(3+x)(1+x^2)} = \frac{A}{3+x} + \frac{Bx+C}{1+x^2}$         |       |
| for all $x \neq -3$                                                    |       |
| (ii) Use part (i) above and the substitution $t = \tan \theta$ to find | 2     |
| $\int \frac{10d\theta}{3 + \tan \theta}$                               |       |

## SECTION A continued

Question 2 (15 marks)

Marks

- |                                                                                                                      |       |                                                                                                                    |   |
|----------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------|---|
| (a)                                                                                                                  | (i)   | Write the complex number $-\sqrt{3} + i$ in modulus-argument form.                                                 | 1 |
|                                                                                                                      | (ii)  | Hence, use de Moivre's Theorem to find $(-\sqrt{3} + i)^{10}$ in the form $a + ib$ , for real values $a$ and $b$ . | 2 |
| (b) Sketch each of the following regions on separate Argand diagrams                                                 |       |                                                                                                                    |   |
|                                                                                                                      | (i)   | $-1 < \operatorname{Re}(z) < 2$ and $0 < \operatorname{Im}(z) < 3$                                                 | 2 |
|                                                                                                                      | (ii)  | $z\bar{z} - (1-i)z - (1+i)\bar{z} < 2$                                                                             | 2 |
|                                                                                                                      | (iii) | $0 < \arg[(1-i)z] < \frac{\pi}{6}$                                                                                 | 2 |
| (c)                                                                                                                  |       |                                                                                                                    |   |
|                                                                                                                      | (i)   | Find the square roots of the complex number $-3 + 4i$                                                              | 2 |
|                                                                                                                      | (ii)  | Find the roots of the quadratic equation $x^2 - (4 - 2i)x + (6 - 8i) = 0$                                          | 2 |
| (d) The locus of a point $P$ , which moves in the complex plane, is represented by the equation $ z - (3 + 4i)  = 5$ |       |                                                                                                                    |   |
|                                                                                                                      | (i)   | Sketch the locus of the point $P$ .                                                                                | 2 |
|                                                                                                                      | (ii)  | Find the modulus of $z$ when $\arg z = \tan^{-1}(\frac{1}{2})$ .                                                   | 2 |

## SECTION B (Use a SEPARATE writing booklet)

Question 3 (15 marks)

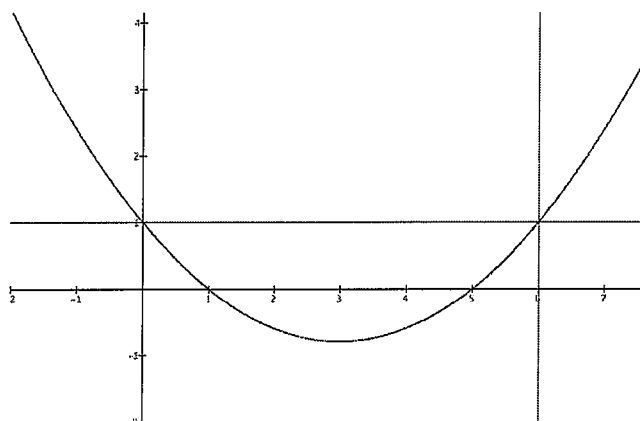
- |                                                                                                                |                                                                                                                                               |                                                                                         |   |
|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---|
| (a)                                                                                                            | Find a cubic equation with roots $\alpha$ , $\beta$ and $\gamma$ such that                                                                    | 3                                                                                       |   |
|                                                                                                                | $\left. \begin{aligned} \alpha\beta\gamma &= 5 \\ \alpha + \beta + \gamma &= 7 \\ \alpha^2 + \beta^2 + \gamma^2 &= 29 \end{aligned} \right\}$ |                                                                                         |   |
| (b) The polynomial $P(x)$ is defined by $P(x) = x^4 + Ax^2 + B$ , where $A$ and $B$ are real positive numbers. |                                                                                                                                               |                                                                                         |   |
|                                                                                                                | (i)                                                                                                                                           | Explain why $P(x)$ has no real zeroes.                                                  | 2 |
|                                                                                                                | (ii)                                                                                                                                          | If two of the zeroes of $P(x)$ are $ib$ and $-id$ where $b$ and $d$ are real show that: | 4 |
|                                                                                                                |                                                                                                                                               | $b^4 + d^4 = A^2 - 2B$                                                                  |   |
| (c) Given that $f(x) = x^3 - 3ax + b$ , where $a$ and $b$ are real numbers then:                               |                                                                                                                                               |                                                                                         |   |
|                                                                                                                | (i)                                                                                                                                           | Show that $y = f(x)$ has turning points if $a > 0$ , and find their coordinates.        | 3 |
|                                                                                                                | (ii)                                                                                                                                          | Show that $f(x)$ has three distinct real zeroes if $b^2 < 4a^3$ .                       | 3 |

## SECTION B continued

Question 4 (15 marks)

Marks

(a)



The sketch above shows the parabola  $y = f(x)$ , where

$$f(x) = \frac{1}{5}(x-1)(x-5).$$

**Without** any use of calculus, draw careful sketches of the following curves, showing all intercepts, asymptotes and turning points.

**NB** The vertex of the parabola is at  $\left(3, -\frac{4}{5}\right)$ .

- |       |                       |   |
|-------|-----------------------|---|
| (i)   | $y = \frac{1}{f(x)}$  | 2 |
| (ii)  | $y = [f(x)]^2$        | 3 |
| (iii) | $y = \tan^{-1}[f(x)]$ | 3 |
| (iv)  | $y = f(\ln x)$        | 3 |

(b) Suppose the function  $f(x) = O(x) + E(x)$ , where  $O(x)$  is odd and  $E(x)$  is even.

- |      |                                                                          |   |
|------|--------------------------------------------------------------------------|---|
| (i)  | By considering $f(-x)$ , find an expression for $O(x)$ in terms of $f$ . | 2 |
| (ii) | Hence write down $O(x)$ when $f(x) = e^x$ .                              | 2 |

SECTION C starts on page 5

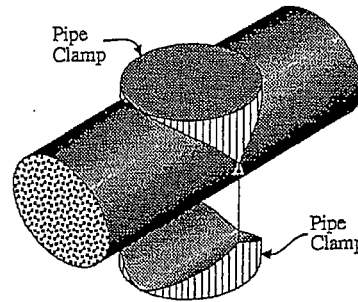
**SECTION C (Use a SEPARATE writing booklet)**

Question 5 (15 marks)

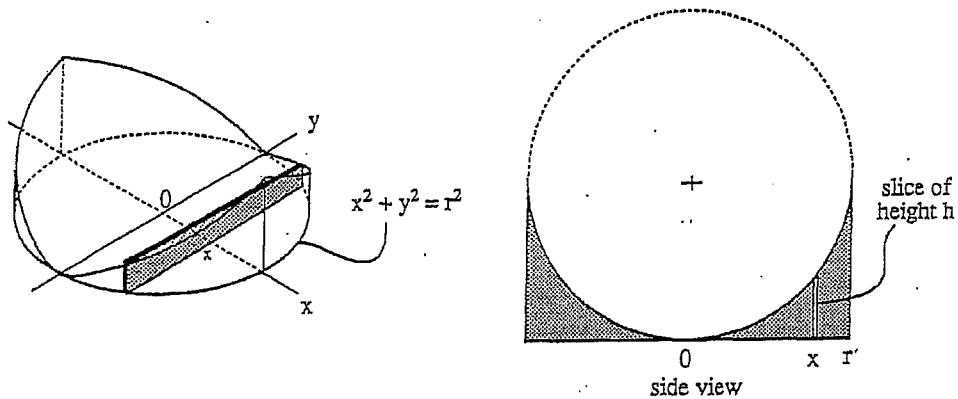
Marks

- (a) A pipe-clamp is made of two identical pieces. Each piece has a circular base of radius  $r$  units and the other face is curved so as to fit flush against the pipe held between the two pieces.

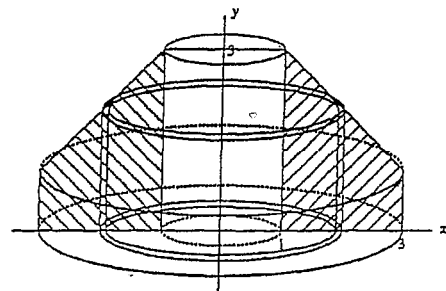
The pipe also has a radius of  $r$  units.



A vertical slice, of thickness  $\Delta x$ , taken  $x$  units from the centre of the base is in the shape of a rectangle with one side in the circular base and of height necessary to reach the cylindrical pipe as shown in the diagram below:



- (i) Show that the height of the slice taken  $x$  units from  $O$  is given by 3
- $$h = r - \sqrt{r^2 - x^2}$$
- (ii) Show that the volume,  $\Delta V$ , of such a slice is given by 3
- $$\Delta V \approx \left[ 2r\sqrt{r^2 - x^2} - 2(r^2 - x^2) \right] \Delta x$$
- (iii) Hence, find by integration, the volume of ONE piece of the pipe-clamp. 3
- (b) (i) Show that the volume,  $\Delta V$ , of a right cylindrical shell of height  $H$ , with inner radius  $r$  and thickness  $\Delta r$  is given by the formula 2
- $$\Delta V = 2\pi r H \Delta r$$
- where  $\Delta r$  is sufficiently small so that  $(\Delta r)^2$  may be neglected.
- (b) (ii) A metal umbrella base is formed by rotating the area enclosed between  $x = 1$ ,  $x = 3$ ,  $y = 0$  and  $y = 4 - x$  about the  $y$ -axis as shown. 4



Using the method of cylindrical shells, find the volume of the umbrella base.

## SECTION C continued

Question 6 (15 marks)

- (a) A point  $T$  moves so that the sum of its distances from the point  $(-2, 0)$  and  $(2, 0)$ , on a Cartesian plane, is 6 units.
- (i) Show that the locus of  $T$  is an ellipse  $\mathcal{E}$  with the equation 2
- $$\frac{x^2}{9} + \frac{y^2}{5} = 1$$
- (ii) Find the equation of the auxiliary circle,  $\mathcal{A}$ , of  $\mathcal{E}$ . 1
- (iii) Find the eccentricity, coordinates of the foci and the equations of the directrices of the ellipse,  $\mathcal{E}$ . 2
- (iv) Draw a neat sketch, showing the ellipse and its auxiliary circle. 1
- (v) A line parallel to the  $y$ -axis meets the positive  $x$ -axis at  $N$  and the curves  $\mathcal{E}$  and  $\mathcal{A}$  at  $P$  and  $Q$  respectively. 1  
 Given the coordinates  $N(3\cos\theta, 0)$ , find the coordinates of  $P$  and  $Q$  (where  $P$  and  $Q$  are in the first quadrant).
- (vi) Find the equations of the tangents at  $P$  and  $Q$ . 2
- (vii) If  $R$  is the point of intersection of the tangents at  $P$  and  $Q$ : 2
- (α) Show that  $R$  lies on the major axis of  $\mathcal{E}$ . 2
- (β) Prove that the product of the lengths  $ON$  and  $OR$  is independent of the positions of  $P$  and  $Q$  on the curves. 2
- (b) Given  $p$  red balls and  $m$  yellow balls, where  $p - m + 1 > 0$ , arranged in a row. 2  
 Show that the number of ways of arranging them so that no two yellow balls appear together is given by:

$${}^{p+1}C_m$$

SECTION D starts on page 7

# SECTION D (Use a SEPARATE writing booklet)

Question 7 (15 marks)

Marks

- (a) (i) Show that  $z^5 + 1 = (z + 1)(z^4 - z^3 + z^2 - z + 1)$  1  
 (ii) If  $z$  is a solution to  $z^5 + 1 = 0$  where  $z \neq -1$ , prove that  $1 + z^2 + z^4 = z + z^3$ . 1  
 (iii) Hence show that  $\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$  3

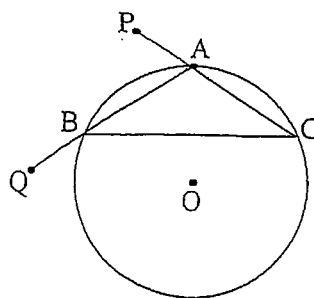
- (b) For integer values of  $k$  where  $k = 0, 1, 2, \dots$  define  $I_k$  as follows:

$$I_k = \int_0^{\frac{\pi}{2}} \cos^k x \, dx$$

- (i) Express  $I_{k+2}$  in terms of  $k$  and  $I_k$ . 2  
 (ii) Hence find an expression for  $I_{2n}$ , where  $n = 0, 1, 2, \dots$  2

- (c) In  $\triangle ABC$ , in the diagram on the right,  $AB = AC$ .

Produce  $CA$  to  $P$  and  $AB$  to  $Q$  so that  
 $AP = BQ$ .



- (i) Show that  $\angle OAP = \angle OBQ$ . 3  
 (ii) Prove that  $A, P, Q$  and  $O$ , the centre of circle  $ABC$ , are concyclic. 3

Question 8 (15 marks)

- (a) A particle is projected **vertically** upwards in a resisting medium where the resistance varies as the square of the velocity and  $k$  is the constant of variation. If the velocity of projection is  $v_0 \tan \alpha$ ,

- (i) Show that the maximum height,  $H$ , reached is given by: 3

$$H = \frac{1}{2k} \ln \left( \frac{g + kv_0^2 \tan^2 \alpha}{g} \right)$$

- (ii) Show that the particle returns to the point of projection with velocity  $v_0 \sin \alpha$  given that  $v_0$  is the terminal velocity. 4

- (iii) Show that the time of ascent is  $\frac{v_0 \alpha}{g}$  3

- (iv) Show that the time of descent is  $\frac{v_0}{g} \ln(\sec \alpha + \tan \alpha)$  2

- (b) Prove by induction that, for all integers  $n$  where  $n > 1$ , that 3

$$\frac{4^n}{n+1} < \frac{(2n)!}{(n!)^2}$$

End of paper

$$1) a) i) \int_{-1}^0 \frac{dx}{\sqrt{3-2x-x^2}}$$

$$= \int_{-1}^0 \frac{dx}{\sqrt{-(x^2+2x+1)+4}}$$

$$= \int_{-1}^0 \frac{dx}{\sqrt{4-(x+1)^2}}$$

$$= \left[ \sin^{-1}\left(\frac{x+1}{2}\right) \right]_{-1}^0$$

$$= \sin^{-1}\left(\frac{1}{2}\right) - \sin^{-1}(0)$$

$$= \frac{\pi}{6}$$

$$ii) \int_0^1 \sqrt{x-x^2} dx$$

$$= \int_0^1 \sqrt{-(x^2-x+\frac{1}{4})+\frac{1}{4}} dx$$

$$= \int_0^1 \sqrt{\frac{1}{4} - (x-\frac{1}{2})^2} dx$$

$$\text{let } x-\frac{1}{2} = \frac{1}{2} \sin \theta$$

$$\frac{dx}{d\theta} = \frac{1}{2} \cos \theta$$

$$dx = \frac{1}{2} \cos \theta d\theta$$

$$\text{when } x=1, \theta = \frac{\pi}{2}$$

$$x=0, \theta = -\frac{\pi}{2}$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{\frac{1}{4} - \frac{1}{4} \sin^2 \theta} \cdot \frac{1}{2} \cos \theta d\theta$$

$$= \frac{1}{4} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{1 - \sin^2 \theta} \cdot \cos \theta d\theta$$

$$= \frac{1}{4} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 \theta d\theta$$

(since even)

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta$$

$$= \frac{1}{4} \int_0^{\frac{\pi}{2}} (1 + \cos 2\theta) d\theta$$

$$= \frac{1}{4} \left[ \theta + \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{4} \left[ \frac{\pi}{2} \right]$$

$$= \frac{\pi}{8}$$

OR

$$\text{let } u = x - \frac{1}{2}$$

$$\frac{du}{dx} = 1$$

$$dx = du$$

$$\text{when } x=1, u = \frac{1}{2}$$

$$x=0, u = -\frac{1}{2}$$

$$= \int_{-\frac{1}{2}}^{\frac{1}{2}} \sqrt{\frac{1}{4} - u^2} du$$

Area of semicircle  
radius  $\frac{1}{2}$ .

$$= \frac{1}{2} \pi \left(\frac{1}{2}\right)^2$$

$$= \frac{\pi}{8}$$

$$b) i) \int \frac{1}{x \ln x} dx$$

$$= \int \frac{\left(\frac{1}{x}\right)}{\ln x} dx$$

$$= \ln(\ln x) + C$$

$$ii) \int x \ln x dx$$

$$\begin{array}{l} u = \ln x \quad v' = x \\ u' = \frac{1}{x} \quad v = \frac{x^2}{2} \end{array}$$

$$= \frac{x^2}{2} \ln x - \int \frac{x}{2} dx$$

$$= \frac{x^2}{2} \ln x - \frac{x^2}{4} + C$$

$$iii) \int \frac{x+1}{x^2+x+1} dx$$

$$= \frac{1}{2} \int \frac{2x+1}{x^2+x+1} dx + \frac{1}{2} \int \frac{dx}{x^2+x+1}$$

$$= \frac{1}{2} \ln(x^2+x+1) + \frac{1}{2} \int \frac{dx}{\left(x^2+x+\frac{1}{4}\right) + \frac{3}{4}}$$

$$= \frac{1}{2} \ln(x^2+x+1) + \frac{1}{2} \int \frac{dx}{\frac{3}{4} + \left(x+\frac{1}{2}\right)^2}$$

$$= \frac{1}{2} \ln(x^2+x+1) + \frac{1}{2} \cdot \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{x+\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) + C$$

$$= \frac{1}{2} \ln(x^2+x+1) + \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + C$$

$$c) \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}} \cos^{-1} x \, dx$$

$$u = \cos^{-1} x \quad v' = 1$$

$$u' = \frac{-1}{\sqrt{1-x^2}} \quad v = x$$

$$= \left[ x \cos^{-1} x \right]_0^{\frac{1}{2}} + \int_0^{\frac{1}{2}} \frac{x}{\sqrt{1-x^2}} \, dx$$

$$\text{let } u = 1-x^2$$

$$\frac{du}{dx} = -2x$$

$$dx = \frac{du}{-2x}$$

$$\text{when } x = \frac{1}{2}, u = \frac{3}{4}$$

$$x = 0, u = 1$$

$$= \frac{1}{2} (\cos^{-1}(\frac{1}{2}) - 0) + \int_1^{\frac{3}{4}} \frac{x}{\sqrt{u}} \cdot \frac{du}{-2x}$$

$$= \frac{1}{2} \cdot \frac{\pi}{3} - \frac{1}{2} \int_1^{\frac{3}{4}} u^{-\frac{1}{2}} du$$

$$= \frac{\pi}{6} - \frac{1}{2} \left[ 2u^{\frac{1}{2}} \right]_1^{\frac{3}{4}}$$

$$= \frac{\pi}{6} - \frac{1}{2} \left[ 2 \cdot \frac{\sqrt{3}}{2} - 2 \right]$$

$$= \frac{\pi}{6} + \frac{2-\sqrt{3}}{2}$$

$$d) i) \frac{10}{(3+x)(1+x^2)} = \frac{A}{3+x} + \frac{Bx+C}{1+x^2}$$

$$10 = A(1+x^2) + (Bx+C)(3+x)$$

$$\text{when } x = -3$$

$$10 = 10A$$

$$A = 1$$

equate coefficients of  $x^2$

$$0 = 1 + B$$

$$B = -1$$

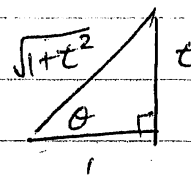
equate constants

$$10 = 1 + 3C$$

$$3C = 9$$

$$C = 3$$

$$ii) \int \frac{10 d\theta}{3 + \tan \theta}$$



$$t = \tan \theta$$

$$\frac{dt}{d\theta} = \sec^2 \theta$$

$$d\theta = \frac{dt}{\sec^2 \theta}$$

$$d\theta = \cos^2 \theta dt$$

$$d\theta = \frac{dt}{1+t^2}$$

$$= \int \frac{10}{3+t} \cdot \frac{dt}{1+t^2} \quad \text{using (i)}$$

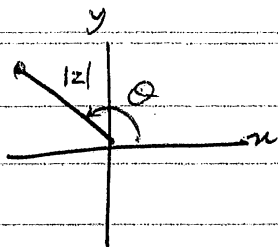
$$= \int \frac{dt}{3+t} + \int \frac{-t+3}{1+t^2} dt$$

$$= \int \frac{dt}{3+t} - \frac{1}{2} \int \frac{2t}{1+t^2} dt + 3 \int \frac{dt}{1+t^2}$$

$$= \ln(3+t) - \frac{1}{2} \ln(1+t^2) + 3 \tan^{-1} t + C$$

$$= \ln(3 + \tan \theta) - \frac{1}{2} \ln(1 + \tan^2 \theta) + 3\theta + C$$

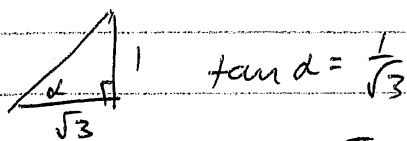
2) a) i)



$$|z| = \sqrt{(-\sqrt{3})^2 + (1)^2}$$

$$= \sqrt{4}$$

$$= 2$$



$$\tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = \frac{\pi}{6}$$

$$\theta = \frac{5\pi}{6}$$

$$\therefore -\sqrt{3} + i = 2 \left( \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)$$

$$\text{ii) } (-\sqrt{3} + i)^{10} = \left[ 2 \left( \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) \right]^{10}$$

$$= 2^{10} \left( \cos \frac{50\pi}{6} + i \sin \frac{50\pi}{6} \right)$$

$$= 1024 \left( \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

$$= 1024 \left( \frac{1}{2} + \frac{\sqrt{3}}{2} i \right)$$

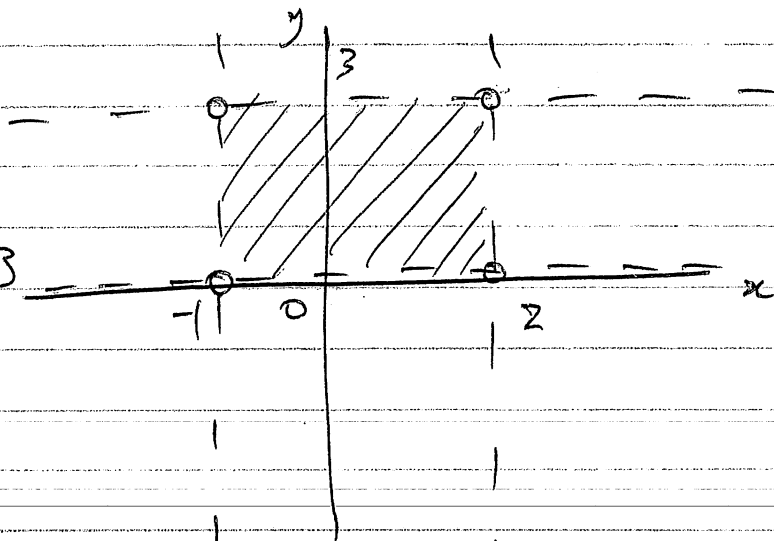
$$= 512 + 512\sqrt{3}i$$

b) i) let  $z = x + iy$

$$-1 < \operatorname{Re}(z) < 2, \quad 0 < \operatorname{Im}(z) < 3$$

$$-1 < \operatorname{Re}(x + iy) < 2, \quad 0 < \operatorname{Im}(x + iy) < 3$$

$$-1 < x < 2, \quad 0 < y < 3$$



ii) let  $z = x + iy$

$$z\bar{z} - (1-i)z - (1+i)\bar{z} < 2$$

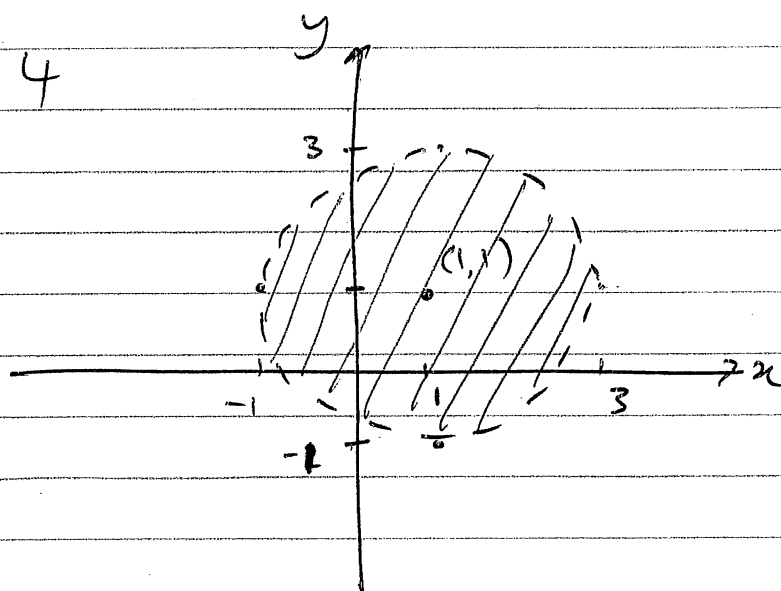
$$(x+iy)(x-iy) - (1-i)(x+iy) - (1+i)(x-iy) < 2$$

$$x^2 + y^2 - (x+iy - ix + y) - (x-iy + ix + y) < 2$$

$$x^2 + y^2 - \cancel{x} - \cancel{iy} + \cancel{ix} - y - \cancel{x} + \cancel{iy} - \cancel{ix} - y < 2$$

$$x^2 - 2x + 1 + y^2 - 2y + 1 < 2 + 1 + 1$$

$$(x-1)^2 + (y-1)^2 < 4$$



iii)  $0 < \arg[(1-i)z] < \frac{\pi}{6}$

\* When two complex numbers are multiplied together

The argument of the resulting complex number is the sum of the individual arguments

$$\arg(1-i) = -\frac{\pi}{4}$$

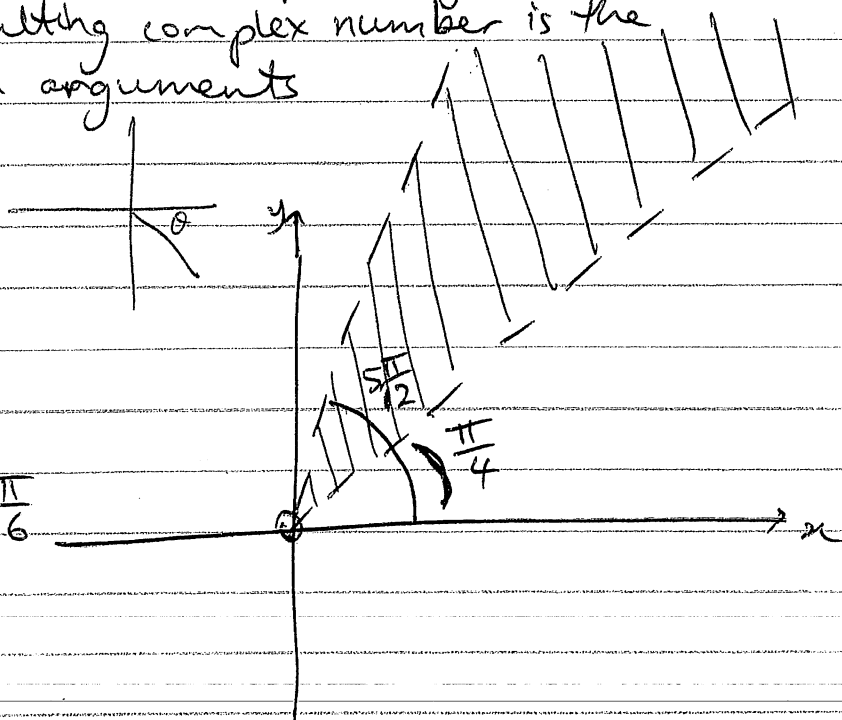
$$\text{let } z = r(\cos \theta + i \sin \theta)$$

$$\arg(z) = \theta$$

$$0 < \arg[(1-i)z] < \frac{\pi}{6}$$

$$0 < \theta - \frac{\pi}{4} < \frac{\pi}{6}$$

$$\frac{\pi}{12} < \theta < \frac{5\pi}{12}$$



c) i) let  $\sqrt{-3+4i} = x+iy$  where  $x$  &  $y$  are real.

$$-3+4i = x^2 - y^2 + 2xyi$$

equating real & imaginary parts

$$x^2 - y^2 = -3 \quad \text{--- (1)}$$

$$2xy = 4 \quad \text{--- (2)}$$

rearrange (2)  $y = \frac{2}{x}$  --- (2a)

sub (2a) into (1)

$$x^2 - \left(\frac{2}{x}\right)^2 = -3$$

$$x^2 - \frac{4}{x^2} = -3$$

$$x^4 - 4 = -3x^2$$

$$x^4 + 3x^2 - 4 = 0$$

$$(x^2 + 4)(x^2 - 1) = 0$$

$$x^2 = -4$$

$$x^2 = 1$$

But  $x^2 \neq -4$  since  
 $x$  is real.

$$x = \pm 1$$

sub into (2a)

when  $x = 1$

$$x = -1$$

$$y = 2$$

$$y = -2$$

$\therefore 1+2i$  &  $-1-2i$  are the square roots of  $-3+4i$ .  
ie  $\pm(1+2i)$  are the square roots of  $-3+4i$

$$ii) \quad x^2 - (4-2i)x + (6-8i) = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{4-2i \pm \sqrt{-(4-2i)^2 - 4(1)(6-8i)}}{2(1)}$$

$$x = \frac{4-2i \pm \sqrt{16-16i-4-24+32i}}{2}$$

$$x = \frac{4-2i \pm \sqrt{-12+16i}}{2}$$

$$x = \frac{4-2i \pm 2\sqrt{-3+4i}}{2}$$

$$x = 2-i \pm \sqrt{-3+4i}$$

from (i)

$$x = 2-i \pm (1+2i)$$

$$x = 2-i + 1+2i \quad \text{or} \quad x = 2-i - 1-2i$$

$$x = 3+i$$

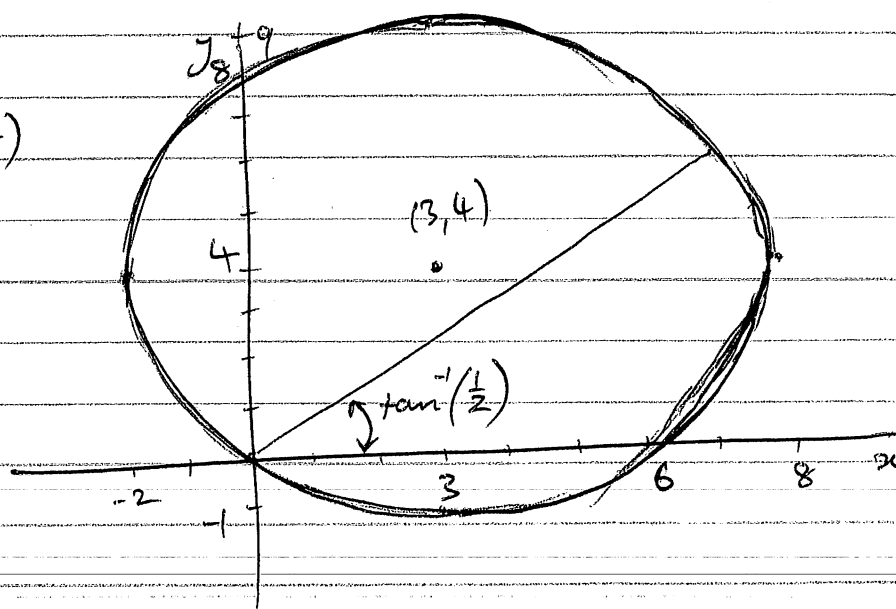
$$x = 1-3i$$

$\therefore$  the roots are  $3+i$  &  $1-3i$ .

$$d) i) \quad |z - (3+4i)| = 5$$

is a circle centre  $(3,4)$

radius 5.



$$i) \theta = \tan^{-1}\left(\frac{1}{2}\right)$$

$$\tan \theta = \frac{1}{2}$$

$$m = \frac{1}{2}$$

$$y = \frac{x}{2} \quad \text{--- (1)}$$

$$(x-3)^2 + (y-4)^2 = 25 \quad \text{--- (2)}$$

sub (1) into (2)

$$(x-3)^2 + \left(\frac{x}{2} - 4\right)^2 = 25$$

$$x^2 - 6x + 9 + \frac{x^2}{4} - 4x + 16 = 25$$

$$4x^2 - 24x + 36 + x^2 - 16x + 64 = 100$$

$$5x^2 - 40x + 100 = 100$$

$$5x(x-8) = 0$$

$$x = 0$$

$$x = 8$$

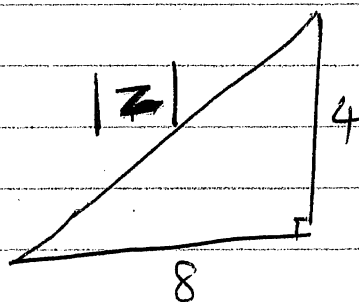
sub into (1)

$$y = 0$$

$$y = 4$$

The line  $y = \frac{x}{2}$  intersects the circle  $(x-3)^2 + (y-4)^2 = 25$  at  $(0, 0)$  and  $(8, 4)$ .

$$\begin{aligned} |z| &= \sqrt{8^2 + 4^2} \\ &= \sqrt{64 + 16} \\ &= \sqrt{80} \\ &= 4\sqrt{5} \end{aligned}$$



QUESTION 3.

a, Let the polynomial be  $x^3 - S_1x^2 + S_2x - S_3 = 0$  (A)

now  $S_1 = \alpha + \beta + \gamma = 7.$

$$S_3 = 5$$

$$S_1^2 = (\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$\text{i.e. } S_1^2 = \alpha^2 + \beta^2 + \gamma^2 + 2S_2.$$

$$49 = 29 + 2S_2$$

$$\therefore S_2 = 10.$$

$\therefore$  (A) becomes  $\boxed{x^3 - 7x^2 + 10x - 5 = 0}$

b, (i)  $P(x) = x^4 + Ax^2 + B$  where A and B are positive

Clearly  $P(x) \geq B \therefore P(x) > 0$  ( $x^4$  and  $Ax^2$  are ~~for~~ non-negative)  
 $\therefore P(x) \neq 0 \therefore$  no real zeros.

NOTE\* There who treated  $P(x)$  as a quadratic struggled to gain marks.

\* Other approaches, commonly used, involved calculus to find that B is the minimum value.

(ii) Since the coefficients are real, by the conjugate root theorem, the four roots are  $\pm ib$  &  $\pm id$ .

$$\begin{aligned} \therefore x^4 + Ax^2 + B &\equiv (x + ib)(x - ib)(x + id)(x - id) \\ &\equiv (x^2 + b^2)(x^2 + d^2) \\ &\equiv x^4 + (b^2 + d^2)x^2 + b^2d^2. \end{aligned}$$

equating  $b^2 + d^2 = A$   
 $b^2d^2 = B$

$$\text{now } b^4 + d^4 = (b^2 + d^2)^2 - 2b^2d^2$$

$$\therefore \underline{b^4 + d^4 = A^2 - 2B.}$$

NOTE. There were several other ways of doing this equation.

Q (1)  $f(x) = x^3 - 3ax + b$

$$f'(x) = 3x^2 - 3a.$$

For turning pts  $f'(x) = 0$

$$\text{i.e. } 3(x^2 - a) = 0$$

$$x^2 = a$$

$$x = \pm\sqrt{a}.$$

$\therefore$  turning pts exist if  $x$  is real

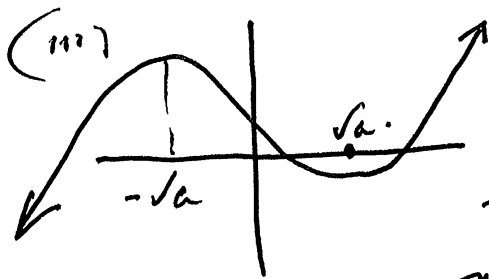
$$\text{i.e. } a > 0$$

(NB. to argue that  $x^2$  is positive therefore  $a > 0$  is not answering the question as asked)

$$\begin{aligned} f(\sqrt{a}) &= a\sqrt{a} - 3a\sqrt{a} + b \\ &= b - 2a\sqrt{a}. \end{aligned}$$

$$\begin{aligned} \& f(-\sqrt{a}) &= -a\sqrt{a} + 3a\sqrt{a} + b \\ &= b + 2a\sqrt{a} \end{aligned}$$

$\therefore$  turning points  
at  $(\sqrt{a}, b - 2a\sqrt{a})$   
and  $(-\sqrt{a}, b + 2a\sqrt{a})$



For three distinct  
real zeros  $f(\sqrt{a})$  and  $f(-\sqrt{a})$   
need to be opposite in sign.

i.e.  $f(\sqrt{a}) \times f(-\sqrt{a}) < 0.$

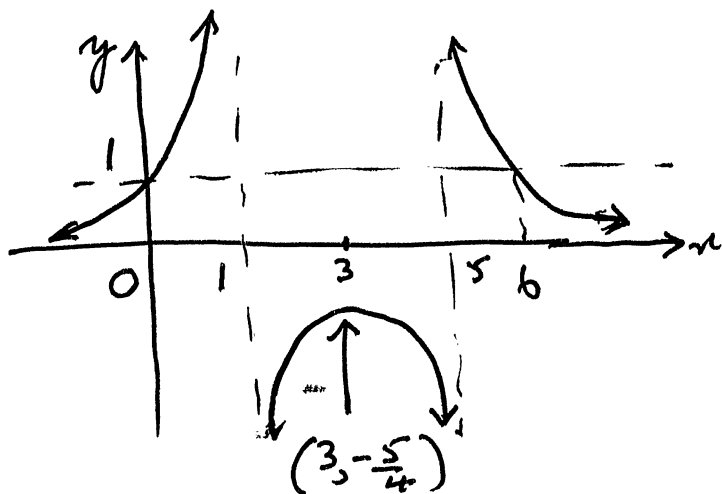
$$(b - 2a\sqrt{a})(b + 2a\sqrt{a}) < 0.$$

$$b^2 - 4a^3 < 0$$

$$\therefore \boxed{b^2 < 4a^3}$$

QUESTION 4.

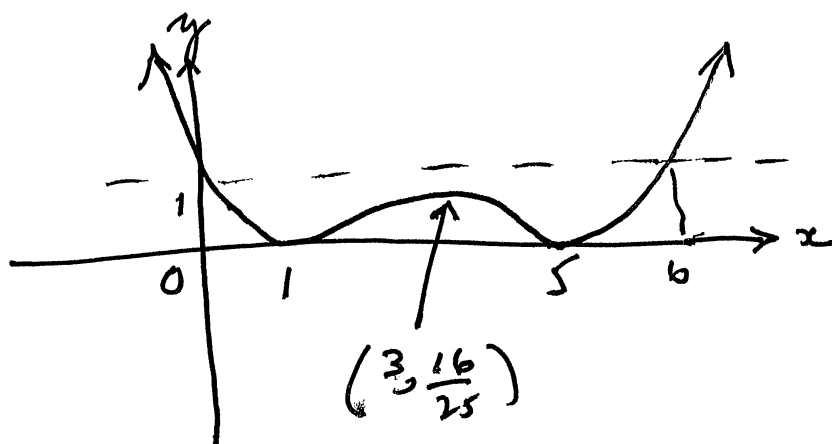
(i)



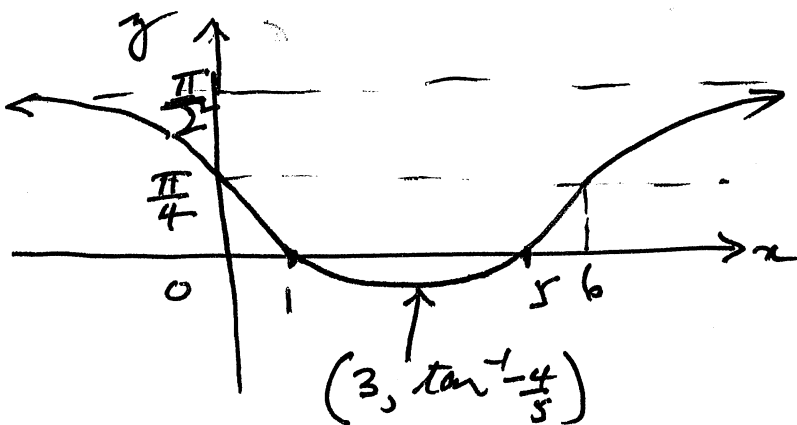
NB

Answers  
required  
zeros, turning  
pts, intercepts  
and asymptotes

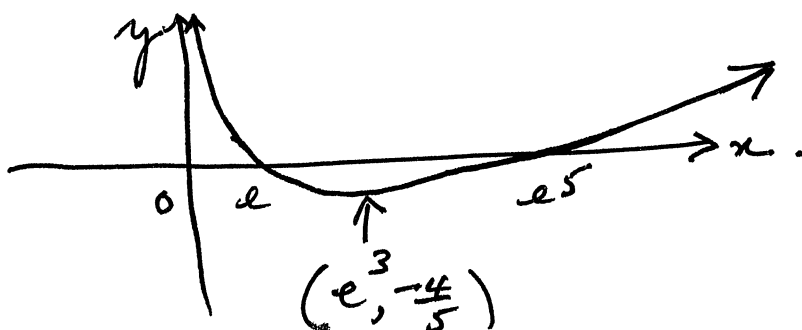
(ii)



(iii)



(iv)



(b) (i) now  $f(x) = O(x) + E(x)$ , — (1)

&  $f(-x) = O(-x) + E(-x)$ , (NB.

i.e.  $f(-x) = -O(x) + E(x)$ , — (2)  $\left. \begin{array}{l} O(-x) = -O(x) \\ \& E(-x) = E(x) \end{array} \right\}$

(1) - (2)

$f(x) - f(-x) = 2 O(x)$

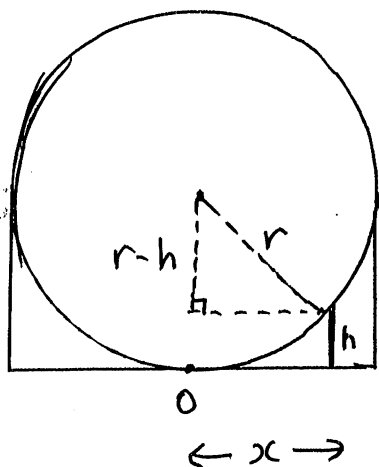
$\therefore \left| O(x) = \frac{f(x) - f(-x)}{2} \right|$

(ii) If  $f(x) = e^x$

$O(x) = \left| \frac{e^x - e^{-x}}{2} \right|$  OR.  $\frac{e^x - 1}{2e^x}$

# QUESTION 5

(a)  
(i)



By Pythagoras' Theorem

$$(r-h)^2 + x^2 = r^2$$

$$\text{i.e. } h^2 - 2rh + x^2 = 0$$

As a quadratic in  $h$

$$\Rightarrow h = \frac{2r \pm \sqrt{4r^2 - 4x^2}}{2}$$

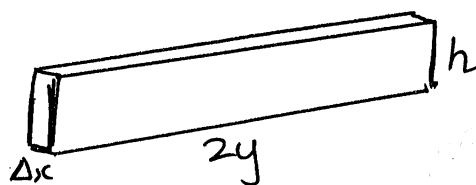
$$h = r \pm \sqrt{r^2 - x^2}$$

Since  $h \leq r$

$$\text{only } h = r - \sqrt{r^2 - x^2}$$

applies

(ii)  $\Delta V = \text{vol. of slice}$



$$\Delta V = A \Delta x$$

$$\Delta V = 2yh \Delta x$$

$$= 2\sqrt{r^2 - x^2} [r - \sqrt{r^2 - x^2}] \Delta x$$

$$\text{i.e. } \Delta V = [2r\sqrt{r^2 - x^2} - 2(r^2 - x^2)] \Delta x$$

(iii)

Vol. of half-clamp  $V$

$$V = 2 \int_0^r (2r\sqrt{r^2 - x^2} - 2r^2 + 2x^2) dx$$

$$= 2 \left\{ 2r \cdot \frac{1}{4} \pi r^2 - [2r^2 x]_0^r + \left[ \frac{2x^3}{3} \right]_0^r \right\}$$

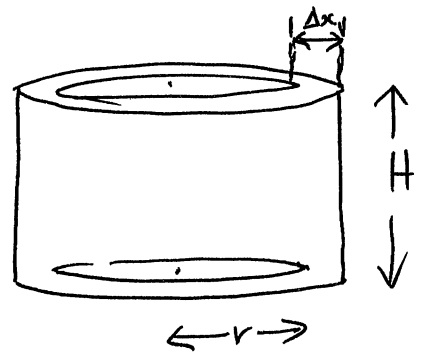
$$= 2 \left\{ \frac{\pi}{2} r^3 - 2r^3 + \frac{2r^3}{3} \right\}$$

$$V = \left\{ \left( \frac{\pi}{2} - \frac{4}{3} \right) r^3 \right\} \times 2 \text{ units}^3$$

$$V = \pi r^3 - \frac{8}{3} r^3$$

$\therefore$  Vol. of whole clamp (one piece)

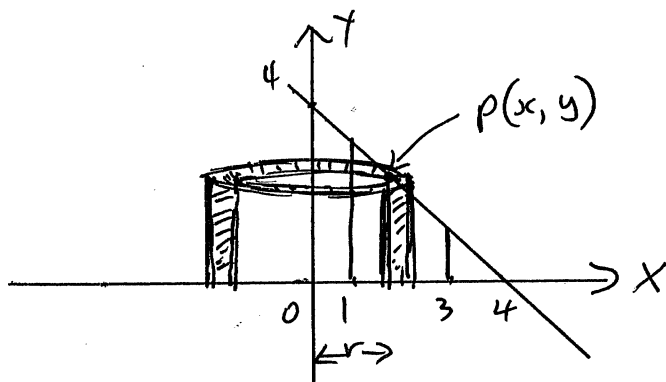
- (5)(b) inner radius of shell =  $r$   
 outer radius of shell =  $r + \Delta r$



(i)  $\Delta V = \text{Vol. of shell}$   
 $= [\pi(r + \Delta r)^2 - \pi r^2] H$   
 $= \pi [r^2 + 2r\Delta r + (\Delta r)^2 - r^2] H$

$\Delta V = 2\pi r H \Delta r$  since  $(\Delta r)^2$  is sufficiently small enough to be neglected.

(ii)



inner radius =  $x$   
 height =  $y$

$$\Delta V = 2\pi r H \Delta r$$

$$\Delta V = 2\pi x y \Delta x \quad \text{where } x = r, \quad y = H$$

$$\therefore V = 2\pi \int_1^3 x y \, dx$$

$$= 2\pi \int_1^3 x(4-x) \, dx$$

$$= 2\pi \left[ 2x^2 - \frac{2x^3}{3} \right]_1^3$$

$$= 2\pi \left[ (18 - 9) - \left( 2 - \frac{1}{3} \right) \right]$$

$$= 2\pi \times \frac{22}{3} = \frac{44\pi}{3} \text{ units}^3$$

## Question 6

(a) let  $S'(-2,0)$   $S(2,0)$   $T(x,y)$   
(i)  $TS' + TS = 6$

$$\sqrt{(x+2)^2 + y^2} + \sqrt{(x-2)^2 + y^2} = 6$$

Squaring, rearranging, squaring

$$\Rightarrow \frac{x^2}{9} + \frac{y^2}{5} = 1$$

(b) Auxiliary circle of radius 3

(ii) centro  $(0,0)$  is

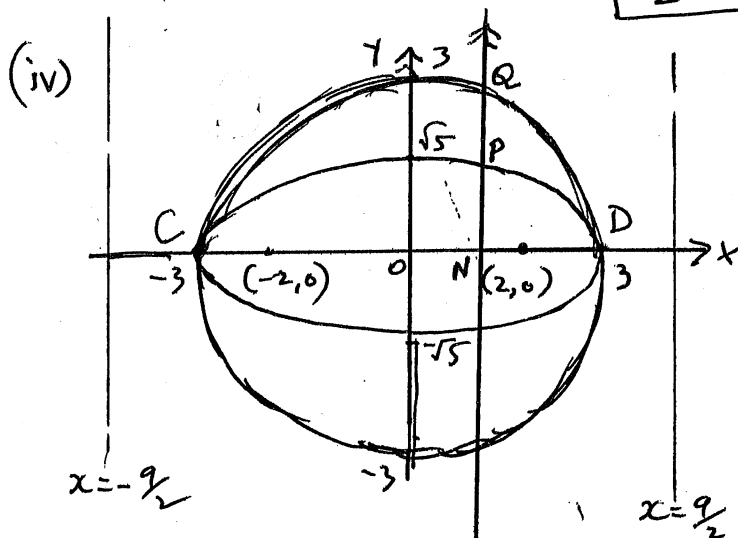
$$x^2 + y^2 = 9$$

(c)  $b^2 = a^2(1 - e^2)$

(iii)  $5 = 9(1 - e^2) \Rightarrow \boxed{e = \frac{2}{3}}$

Foci  $(\pm ae, 0)$  i.e.  $(\pm 2, 0)$

Directrices  $x = \pm \frac{a}{e} \Rightarrow \boxed{\pm \frac{9}{2} = x}$



(v)  $N(3\cos\theta, 0)$

$P(3\cos\theta, \sqrt{5}\sin\theta)$

$Q(3\cos\theta, 3\sin\theta)$

(vi)

Tangent at P is

$$\frac{x\cos\theta}{3} + \frac{y\sqrt{5}\sin\theta}{5\sin\theta} = 1 \quad \textcircled{A}$$

Tangent at Q

grad. is  $2x + 2yy' = 0$

$$y' = -\frac{x}{y}$$

at Q grad is  $-\frac{3\cos\theta}{3\sin\theta} = -\cot\theta$

Eq<sup>n</sup> is  $y - 3\sin\theta = -\cot\theta(x - 3\cos\theta)$

$\Rightarrow y\sin\theta + x\cos\theta = 3 \quad \textcircled{B}$

(vii)  $\Rightarrow$  Solving  $\textcircled{A}$  and  $\textcircled{B}$

$$5x\cos\theta + 3\sqrt{5}y\sin\theta = 15 \quad \textcircled{A}$$

$$x\cos\theta + y\sin\theta = 3 \quad \textcircled{B}$$

$$5 \times \textcircled{B} \Rightarrow 5x\cos\theta + 5y\sin\theta = 15 \quad \textcircled{C}$$

$$\textcircled{A} - \textcircled{C} \Rightarrow (3\sqrt{5} - 5)y\sin\theta = 0$$

$$3\sqrt{5} - 5 \neq 0, \sin\theta \neq 0 \text{ except at } C, D$$

$$\Rightarrow \boxed{y = 0}$$

$$\boxed{x = \frac{3}{\cos\theta}} \quad \cos\theta \neq 0$$

$\therefore \forall x$  point R  $(\frac{3}{\cos\theta}, 0)$  lies on the x-axis.

(p)

$$\text{ON} \cdot \text{OR} = 3\cos\theta \cdot \frac{3}{\cos\theta} = 9$$

$\therefore$  independent of P and Q.

7. (a) (i) Show that
- $z^5 + 1 = (z + 1)(z^4 - z^3 + z^2 - z + 1)$
- .

1

**Solution:**

$$\begin{aligned}
 \text{R.H.S.} &= z(z^4 - z^3 + z^2 - z + 1) + (z^4 - z^3 + z^2 - z + 1) \\
 &= (z^5 - z^4 + z^3 - z^2 + z) + (z^4 - z^3 + z^2 - z + 1) \\
 &= z^5 + 1 \\
 &= \text{L.H.S.}
 \end{aligned}$$

- (ii) If
- $z$
- is a solution to
- $z^5 + 1 = 0$
- where
- $z \neq -1$
- , prove that
- $1 + z^2 + z^4 = z + z^3$
- .

1

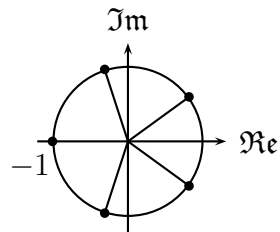
**Solution:**

$$\begin{aligned}
 (z + 1)(z^4 - z^3 + z^2 - z + 1) &= 0 \quad \text{but } z \neq -1, \\
 \therefore z^4 - z^3 + z^2 - z + 1 &= 0. \\
 \text{Hence } 1 + z^2 + z^4 &= z + z^3.
 \end{aligned}$$

- (iii) Hence show that
- $\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$
- .

3

**Solution:**



From the diagram

$$\text{if } z^5 = -1$$

$$z = \text{cis } \pm \frac{\pi}{5}, \text{cis } \pm \frac{3\pi}{5}, -1$$

Method 1: We take  $z = \text{cis } \frac{\pi}{5}$ .

$$1 + z^2 + z^4 = z + z^3 \text{ from (ii),}$$

$$\frac{1}{z^2} + 1 + z^2 = \frac{1}{z} + z,$$

$$2 \cos \frac{2\pi}{5} + 1 = 2 \cos \frac{\pi}{5},$$

$$2 \cos \frac{\pi}{5} - 2 \cos \frac{2\pi}{5} = 1,$$

$$2 \cos \frac{\pi}{5} + 2 \cos \frac{3\pi}{5} = 1,$$

$$\therefore \cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}.$$

Method 2: We consider the roots of  $z^4 - z^3 + z^2 - z + 1 = 0$  from (ii), taken one-at-a-time,

$$\text{cis } \frac{\pi}{5} + \text{cis } \frac{-\pi}{5} + \text{cis } \frac{3\pi}{5} + \text{cis } \frac{-3\pi}{5} = 1.$$

$$\text{But } z + \bar{z} = 2\Re(z),$$

$$\text{so } \text{cis } \frac{\pi}{5} + \text{cis } \frac{-\pi}{5} = 2 \cos \frac{\pi}{5} \text{ etc.}$$

$$\therefore 2 \cos \frac{\pi}{5} + 2 \cos \frac{3\pi}{5} = 1,$$

$$\text{and } \cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}.$$

(b) For integer values of  $k$  where  $k = 0, 1, 2, \dots$  define  $I_k$  as follows:

$$I_k = \int_0^{\frac{\pi}{2}} \cos^k x \, dx$$

(i) Express  $I_{k+2}$  in terms of  $k$  and  $I_k$ .

2

**Solution:**

$$\begin{aligned} u &= \cos^{k+1} x & v' &= \cos x \\ u' &= (k+1)(-\sin x) \cos^k x & v &= \sin x \end{aligned}$$

$$\begin{aligned} I_{k+2} &= \int_0^{\frac{\pi}{2}} \cos^{k+1} x \cdot \cos x \, dx, \\ &= \left[ \sin x \cos^{k+1} x \right]_0^{\frac{\pi}{2}} + (k+1) \int_0^{\frac{\pi}{2}} \sin^2 x \cdot \cos^k x \, dx, \\ &= 0 + (k+1) \int_0^{\frac{\pi}{2}} (\cos^k x - \cos^{k+2} x) \, dx, \\ &= (k+1)I_k - (k+1)I_{k+2}, \\ (k+2)I_{k+2} &= (k+1)I_k, \\ I_{k+2} &= \left( \frac{k+1}{k+2} \right) I_k. \end{aligned}$$

(ii) Hence find an expression for  $I_{2n}$ , where  $n = 0, 1, 2, \dots$

2

**Solution:**  $I_0 = \int_0^{\frac{\pi}{2}} dx = I_{2 \times 0} \quad \text{i.e. } n = 0,$

$$= \frac{\pi}{2}.$$

$$I_2 = I_{0+2} = I_{2 \times 1} \quad \text{i.e. } n = 1,$$

$$= \frac{1}{2} \cdot I_0,$$

$$= \frac{1}{2} \cdot \frac{\pi}{2}.$$

$$I_4 = I_{2+2} = I_{2 \times 2} \quad \text{i.e. } n = 2,$$

$$= \frac{3}{4} \cdot I_2,$$

$$= \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}.$$

$$I_6 = I_{4+2} = I_{2 \times 3} \quad \text{i.e. } n = 3,$$

$$= \frac{5}{6} \cdot I_4,$$

$$= \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2},$$

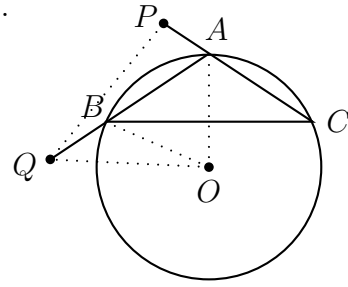
$$= \frac{6}{6} \cdot \frac{5}{6} \cdot \frac{4}{4} \cdot \frac{3}{4} \cdot \frac{2}{2} \cdot \frac{1}{2} \cdot \frac{\pi}{2},$$

$$= \frac{1}{2^6} \cdot \frac{6!}{(3!)^2} \cdot \frac{\pi}{2}.$$

$$I_{2n} = \frac{1}{2^{2n}} \cdot \frac{(2n)!}{(n!)^2} \cdot \frac{\pi}{2}.$$

(c) In  $\triangle ABC$ , in the diagram on the right,  $AB = AC$ .

Produce  $CA$  to  $P$  and  $AB$  to  $Q$  so that  $AP = BQ$ .



(i) Show that  $\angle OAP = \angle OBQ$ .

3

**Solution:** In  $\triangle AOC$ ,  $\triangle AOB$ ,  
 $AC = AB$  (data),  
 $OA = OB = OC$  (equal radii),  
 $\therefore \triangle AOC \equiv \triangle AOB$  (SSS),  
 $\angle OAC = \angle OBA$  (corresp.  $\angle$ s in congruent  $\triangle$ s),  
 $\angle OAP + \angle OAC = 180^\circ (= \angle PAC)$ ,  
 $\angle OBQ + \angle OBA = 180^\circ (= \angle ABQ)$ ,  
 $\therefore \angle OAP = \angle OBQ$ .

(ii) Prove that  $A$ ,  $P$ ,  $Q$ , and  $O$ , the centre of the circle,  $ABC$  are concyclic.

3

**Solution:** In  $\triangle OAP$ ,  $\triangle OBQ$ ,  
 $AP = BQ$  (data),  
 $\angle OAP = \angle OBQ$  (shown above),  
 $OA = OB$  (equal radii),  
 $\therefore \triangle OAP \equiv \triangle OBQ$  (SAS),  
 $\angle APO = \angle BQO$  (corresp.  $\angle$ s in congruent  $\triangle$ s),  
 $\angle BQO = \angle AQO$  (common),  
 $\therefore \angle APO = \angle AQO$   
 $\therefore A$ ,  $P$ ,  $Q$ , and  $O$  are concyclic  
(equal angles subtended by the chord  $AO$ ).

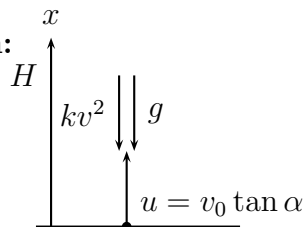
8. (a) A particle is projected **vertically** upwards in a resisting medium where the resistance varies as the square of the velocity and  $k$  is the constant of variation. If the velocity of projection is  $v_0 \tan \alpha$ ,

(i) Show that the maximum height,  $H$ , reached is given by:

3

$$H = \frac{1}{2k} \ln \left( \frac{g + kv_0^2 \tan^2 \alpha}{g} \right)$$

**Solution:**



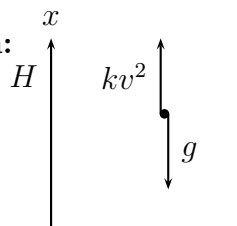
$$\begin{aligned} \ddot{x} &= -kv^2 - g \\ v \frac{dv}{dx} &= -(kv^2 + g) \\ \frac{dv}{dx} &= -\frac{kv^2 + g}{v} \\ \int_0^H dx &= -\frac{1}{2k} \int_u^v \frac{2kv dv}{kv^2 + g} \\ 0 - H &= -\frac{1}{2k} \left[ \ln(kv^2 + g) \right]_u^0 \end{aligned}$$

Now, replacing  $u$  with  $v_0 \tan \alpha$ ,  $H = \frac{1}{2k} \ln \left( \frac{g + kv_0^2 \tan^2 \alpha}{g} \right)$ .

- (ii) Show that the particle returns to the point of projection with velocity  $v_0 \sin \alpha$  given that  $v_0$  is the terminal velocity.

4

**Solution:**



$$\begin{aligned} \ddot{x} &= kv^2 - g, \\ v \frac{dv}{dx} &= kv^2 - g. \\ \text{At terminal velocity, } \ddot{x} &= 0, \\ \therefore kv_0^2 &= g, \\ \text{i.e. } v_0^2 &= \frac{g}{k} \text{ or } v_0 = \sqrt{\frac{g}{k}}. \end{aligned}$$

$$\begin{aligned} \int_H^0 dx &= -\frac{1}{2k} \int_0^{-v} \frac{2kv dv}{kv^2 - g}, \\ 0 - H &= \frac{1}{2k} \left[ \ln(kv^2 - g) \right]_0^{-v}, \\ -H &= \frac{1}{2k} \ln \left( \frac{kv^2 - g}{-g} \right), \end{aligned}$$

Now, equating  $H$ s,

$$\begin{aligned} \frac{1}{2k} \ln \left( \frac{g + kv_0^2 \tan^2 \alpha}{g} \right) &= \frac{1}{2k} \ln \left( \frac{g}{g - kv^2} \right), \\ \text{So, } \frac{g + kv_0^2 \tan^2 \alpha}{g} &= \frac{g}{g - kv^2}, \end{aligned}$$

$$\begin{aligned}
1 + \frac{kv_0^2}{g} \cdot \tan^2 \alpha &= \frac{1}{1 - \frac{k}{g}v^2}, \\
1 + \tan^2 \alpha &= \frac{1}{1 - \frac{v^2}{v_0^2}}, \\
\frac{\sec^2 \alpha}{v_0^2} &= \frac{1}{v_0^2 - v^2}, \\
v_0^2 - v^2 &= v_0^2 \cos^2 \alpha, \\
&= v_0^2 - v_0^2 \sin^2 \alpha, \\
v^2 &= v_0^2 \sin^2 \alpha, \\
v &= v_0 \sin \alpha.
\end{aligned}$$

(iii) Show that the time of ascent is  $\frac{v_0 \alpha}{g}$

3

**Solution:** From part (i),

$$\begin{aligned}
\ddot{x} &= -(g + kv^2), \\
\frac{dv}{dt} &= -(g + kv^2). \\
\int_0^T dt &= - \int_u^0 \frac{dv}{g + kv^2}, \\
&= \frac{1}{k} \int_0^{v_0 \tan \alpha} \frac{dv}{v_0^2 + v^2}, \text{ using } \frac{g}{k} = v_0^2 \text{ from (ii),} \\
T &= \frac{1}{k} \cdot \frac{1}{v_0} \left[ \tan^{-1} \frac{v}{v_0} \right]_0^{v_0 \tan \alpha}, \\
&= \frac{1}{kv_0} \tan^{-1} \tan \alpha,
\end{aligned}$$

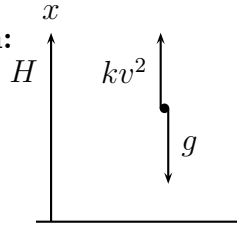
$\therefore$  time of ascent,

$$\begin{aligned}
T &= \frac{\alpha}{kv_0}, \quad \text{but } k = \frac{g}{v_0^2}, \\
&= \frac{\alpha}{v_0} \cdot \frac{v_0^2}{g}, \\
&= \frac{v_0 \alpha}{g}.
\end{aligned}$$

(iv) Show that the time of descent is  $\frac{v_0}{g} \ln(\sec \alpha + \tan \alpha)$ .

2

**Solution:**



$$\begin{aligned}\ddot{x} &= kv^2 - g, \\ \frac{dv}{dt} &= kv^2 - g, \\ \int_0^T dt &= \int_0^{-v_0 \sin \alpha} \frac{dv}{kv^2 - g}, \\ &= \frac{1}{k} \int_0^{-v_0 \sin \alpha} \frac{dv}{v^2 - v_0^2}.\end{aligned}$$

$$\begin{aligned}\frac{1}{v^2 - v_0^2} &\equiv \frac{A}{v + v_0} + \frac{B}{v - v_0}, \\ 1 &\equiv A(v - v_0) + B(v + v_0).\end{aligned}$$

$$\text{Put } v = v_0, \quad B = \frac{1}{2v_0},$$

$$\text{put } v = -v_0, \quad A = -\frac{1}{2v_0}.$$

$$\begin{aligned}\text{Then, } T &= \frac{1}{2kv_0} \int_0^{-v_0 \sin \alpha} \left( \frac{1}{v - v_0} - \frac{1}{v + v_0} \right) dv, \\ &= \frac{1}{2kv_0} \left[ \ln(v - v_0) - \ln(v + v_0) \right]_0^{-v_0 \sin \alpha}, \\ &= \frac{1}{2kv_0} \left\{ \ln \left( \frac{-v_0 \sin \alpha - v_0}{-v_0} \right) - \ln \left( \frac{v_0 - v_0 \sin \alpha}{v_0} \right) \right\}, \\ &= \frac{1}{2kv_0} \ln \left\{ \left( \frac{\sin \alpha + 1}{1 - \sin \alpha} \right) \cdot \left( \frac{1 + \sin \alpha}{1 + \sin \alpha} \right) \right\}, \\ &= \frac{1}{2kv_0} \ln \left\{ \frac{(\sin \alpha + 1)^2}{\cos^2 \alpha} \right\}, \\ &= \frac{1}{kv_0} \ln \sqrt{(\sec \alpha + \tan \alpha)^2}, \\ &= \frac{1}{kv_0} \ln(\sec \alpha + \tan \alpha).\end{aligned}$$

(b) Prove by induction that, for all integers  $n$  where  $n > 1$ , that

3

$$\frac{4^n}{n+1} < \frac{(2n)!}{(n!)^2}$$

**Solution:** Test for  $n = 2$ :

$$\begin{aligned} \text{L.H.S.} &= \frac{4^2}{3}, & \text{R.H.S.} &= \frac{4!}{(2!)^2}, \\ &= 5\frac{1}{3}. & &= 6. \end{aligned}$$

$\therefore$  True for  $n = 2$ .

Now, assume true for  $n = k \geq 2$ ,

$$i.e. \quad \frac{4^k}{k+1} < \frac{(2k)!}{(k!)^2}$$

Then test for  $n = k + 1$ ,

$$i.e. \quad \frac{4^{k+1}}{k+2} < \frac{(2k+2)!}{((k+1)!)^2}$$

$$\begin{aligned} \text{L.H.S.} &= \frac{4 \cdot 4^k}{k+1} \times \frac{k+1}{k+2}, \\ &< \frac{(2k)!}{(k!)^2} \times \frac{4(k+1)}{k+2} \quad \text{by the assumption,} \\ &< \frac{(2k+2)!}{((k+1)!)^2} \times \frac{(k+1)^2}{(2k+2)(2k+1)} \times \frac{4(k+1)}{(k+2)}, \\ &< \frac{(2k+2)!}{((k+1)!)^2} \times \frac{(2k^2+4k+2)}{(2k^2+5k+2)}, \\ &< \frac{(2k+2)!}{((k+1)!)^2} \times \left(1 - \frac{k}{2k^2+5k+2}\right), \\ &< \frac{(2k+2)!}{((k+1)!)^2} \quad \text{as } \left(1 - \frac{k}{2k^2+5k+2}\right) < 1. \end{aligned}$$

$$\begin{aligned} \text{Alternatively, R.H.S.} &= \frac{(2k)!}{(k!)^2} \times \frac{(2k+2)(2k+1)}{(k+1)^2}, \\ &> \frac{4^k}{k+1} \times \frac{2(2k+1)}{(k+1)} \quad \text{by the assumption,} \\ &> \frac{4^{k+1}}{k+2} \times \frac{(k+2)}{4(k+1)} \times \frac{2(2k+1)}{(k+1)}, \\ &> \frac{4^{k+1}}{k+2} \times \frac{(2k^2+5k+2)}{(2k^2+4k+2)}, \\ &> \frac{4^{k+1}}{k+2} \times \left(1 + \frac{k}{2k^2+4k+2}\right), \\ &> \frac{4^{k+1}}{k+2} \quad \text{as } \left(1 + \frac{k}{2k^2+4k+2}\right) > 1. \end{aligned}$$

$\therefore$  The statement is true for  $n = k + 1$  if true for  $n = k$ .

As true for  $n = 2$ , so true for  $n = 3, 4, \dots$  and so on for all natural numbers  $n > 1$ .